

2 From Representations to Decisions

Exercises

A note before you start. Try each question yourself before turning to generative AI. Thinking through a problem, struggling with it for a while, and attempting your own answer are part of the learning process. Reading a polished AI-generated solution can easily create the impression that you understand something before you can explain or apply it yourself. If you get stuck, use generative AI to ask for a hint, a simpler question, or feedback on your reasoning rather than immediately asking for the solution. Make AI help you think, rather than do the thinking for you.

The exercises below emphasize explanation, counterexamples, modeling choices, and connections between the main ideas of Lecture 2. Justify each answer rather than reporting only a formula or numerical result.

1. Representation and unavoidable ambiguity

Two physically different network events, z_A and z_B , are mapped to the same recorded feature vector $\mathbf{x} = g(z_A) = g(z_B)$. Event z_A is benign and event z_B is malicious.

- (a) Explain why no deterministic classifier using only $\mathbf{x}$ can classify both events correctly.
- (b) Suppose that among events represented by $\mathbf{x}$, 30% are malicious. Determine the Bayes decision and its conditional error under equal classification costs.
- (c) Propose one additional feature that might separate the events. Explain when that feature would constitute legitimate information and when it would constitute leakage.
- (d) Can adding irrelevant features leave the Bayes error unchanged while making learning from finite data harder? Explain.

2. The loss changes the meaning of the best prediction

For a fixed input $\mathbf{x}$, the response time Y is continuous and has the conditional exponential density

$$p(y \mid \mathbf{x}) = \frac{1}{4}e^{-y/4}, \quad y \geq 0,$$

with conditional distribution function

$$F(y \mid \mathbf{x}) = 1 - e^{-y/4}.$$

The point prediction is an action $a \in [0, \infty)$.

- (a) Find an optimal point prediction under squared loss and under absolute loss. Explain why the answers differ.

- (b) Now use the asymmetric loss

$$\ell(y, a) = \begin{cases} 4(y - a), & a < y, \\ a - y, & a \geq y. \end{cases}$$

Determine an optimal decision and interpret the asymmetry.

- (c) A model outputs only the conditional mean. Explain why this output may be insufficient when the application later changes to the asymmetric loss. What richer quantity could the model estimate instead?

3. Classification with a reject option

A security system estimates $p(\mathbf{x}) = P(Y = 1 \mid \mathbf{X} = \mathbf{x})$, where class 1 denotes an attack. It may declare “attack,” declare “benign,” or defer the case to a human. A false positive costs 1, a false negative costs 9, and deferral always costs 0.6. Correct automatic decisions have zero cost.

- (a) Write the conditional risk of each of the three actions.
- (b) Derive the intervals of $p(\mathbf{x})$ for which each action is optimal.
- (c) Explain why one threshold is no longer sufficient.
- (d) Describe how the decision regions change if human review becomes more expensive.

4. Training loss and decision performance can disagree

Two binary classifiers are evaluated on four examples. The table gives the probability assigned to the correct class.

	1	2	3	4
Model A	0.60	0.60	0.60	0.60
Model B	0.99	0.99	0.99	0.20

- (a) Compare the models using zero-one loss with threshold 0.5.
- (b) Compare their average cross-entropy losses.
- (c) Explain why the two criteria can rank the models differently.
- (d) Does this example show that cross-entropy is an inappropriate training loss? Give a careful argument involving probability estimation and the distinction between training and decision losses.

5. A nearest neighbor created by the units

Consider a query representing a person with income 50,000 and age 30:

$$\mathbf{x} = (50,000, 30).$$

Two training examples are

$$\mathbf{x}_A = (51,000, 60), \quad \mathbf{x}_B = (60,000, 31).$$

- (a) Determine the nearest example using raw Euclidean distance.
- (b) Standardize income using a scale of 10,000 and age using a scale of 10. Determine the nearest example again.
- (c) Explain what changed mathematically and what did not change about the underlying people.
- (d) Suppose location is represented by one-hot encoding. What assumption does Euclidean distance make about the distances between different locations? Propose a domain-aware alternative.

6. Local neighborhoods in high dimension

Assume n points are sampled independently and uniformly from the unit d -ball. A query is located at the center, and a useful neighbor must be within Euclidean distance $r = 0.1$. Let M_d be the number of sampled points in this neighborhood.

- (a) Derive the distribution of M_d , its expectation, and $P(M_d \geq 1)$.
- (b) For $n = 1000$, find the first dimension for which $P(M_d \geq 1) < 0.5$.
- (c) Derive the radius r_d required to obtain $P(M_d \geq 1) = 0.95$ for fixed n . Describe what happens to r_d as d increases.
- (d) Which assumptions in this calculation are favorable to KNN? Discuss the query location, uniformity, and irrelevant dimensions.

7. Perfect interpolation does not determine behavior elsewhere

The training data are

$$(-1, 1), \quad (0, 0), \quad (1, 1).$$

Consider

$$f_1(x) = x^2, \quad f_{2,\alpha}(x) = x^2 + \alpha(x+1)x(x-1),$$

where $\alpha \in \mathbb{R}$.

- (a) Show that all these functions have zero squared training error.
- (b) Compare their predictions at $x = 1/2$.
- (c) Explain why empirical risk minimization alone cannot choose among them.
- (d) Identify the extra assumptions introduced by preferring the quadratic model, applying regularization, or using validation data.

8. Are parameters and hyperparameters intrinsic labels?

In one learning system, polynomial degree p is selected by validation and the coefficients $\theta_0, \dots, \theta_p$ are fitted on the training data. In a second system, a degree-20 polynomial is always used, but an L_1 penalty encourages most coefficients to become zero.

- (a) Identify the parameters and hyperparameters in each system.
- (b) Explain how both systems can produce models with the same effective degree through different learning procedures.
- (c) Is “polynomial degree” always a hyperparameter? Defend your answer.
- (d) What information must remain outside parameter fitting to compare the two procedures fairly?

9. Several empirical-risk minimizers

The input population contains regions A , B , and C . Two classifiers h_1 and h_2 make identical predictions on A and B but opposite predictions on C . A finite training sample contains no observations from C , although $P(\mathbf{X} \in C) > 0$.

- (a) Explain why training loss cannot distinguish h_1 from h_2 .
- (b) Can one infer from the tie that the classifiers have equal population risk? Construct two possible data-generating distributions to support your answer.
- (c) What role does the learning algorithm’s tie-breaking rule play?

- (d) Suggest three different ways to obtain evidence about behavior in region C .

10. Audit a misleading security experiment

A team collects network flows. Each attack produces hundreds of nearly duplicate flows. The team standardizes all records before splitting them, randomly assigns individual rows to training and test sets, tries 500 model configurations, and reports the configuration with the best test accuracy. The selected model uses a threshold of 0.5, although a missed attack costs 100 times more than a false alarm.

- (a) Identify at least four reasons why the reported accuracy may give a misleading picture of deployment performance.
- (b) Redesign the data split and preprocessing pipeline.
- (c) Separate the roles of training loss, validation criterion, test metric, and deployment decision loss.
- (d) Propose a final set of reported results that would allow a security analyst to judge the system responsibly.